

Niraj Kumar Sinha

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ACADEMIC & TECHNICAL EXPERTISE

Programming & Tools

Python, Basic Java, Power BI, MS Excel, PowerPoint, VS Code, IntelliJ IDEA, Jupyter Notebook

Libraries & Frameworks

Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn.

Core Subjects

Machine Learning, Data Science, Artificial Intelligence, Deep Learning, Computer Vision, DBMS, Data Structures, Operating System

Teaching & Professional Skills

Curriculum Delivery • Concept Simplification • Student Mentoring • Problem-Solving • Critical Thinking • Time Management • Academic Communication • Risk Analysis

TEACHING & INDUSTRY EXPERIENCE

Assistant Professor

FACE PREP (Sanskriti University – 9 Months & Rajalakshmi Engineering College – 2 Months)s | 11 Months

- Delivered **hands-on training sessions** on **Python, Data Analysis, and Machine Learning fundamentals** to undergraduate learners.
- Mentored students on **real-world datasets**, improving conceptual clarity and practical coding confidence.
- Designed **practice problems, mini-projects, and assessments**, leading to measurable improvement in learner performance and engagement.
- Recognized for **clear explanation of complex ML concepts** using practical examples and real-life analogies.

EDUCATION

Chandigarh University Mohali	Mohali, Punjab
Master of Engineering – Artificial Intelligence, CGPA -7.42	Aug 2023 – 2025
Vinoba Bhawe University, Hazaribagh	Hazaribagh, Jharkhand
Bachelor of Education, Percentage – 79%	August 2021 – June 2023
Ranchi University Ranchi	Ranchi Jharkhand
Master of Science – Mathematics, CGPA – 8.69	July 2019 - June 2021

RESEARCH PROJECTS

An Ensemble Machine Learning Approach for Diabetes Detection

July 2024 – Dec 2024

- Designed and implemented an **ensemble Random Forest model**, achieving **96.8% prediction accuracy**, outperforming standard benchmarks.
- Improved diabetes prediction reliability by **26%**, supporting early diagnosis and healthcare decision-making.
- Demonstrated strong research orientation by aligning methodology with peer-reviewed standards.

Credit Card Fraud Detection using Machine Learning

Sept 2024 – Oct 2024

- Developed a **Logistic Regression–based fraud detection system** with **96% accuracy**.
- Reduced **false positives by 16%** through feature engineering and hyperparameter optimization.
- Enhanced model efficiency by **23%**, ensuring faster and more reliable fraud identification.

PUBLICATIONS

- “Prediction of Diabetes using Machine Learning Approach” in IEEE Conference. [Link](#)
- “A Survey on Diabetes Mellitus Using Machine Learning Classifiers” in IEEE Conference. [Link](#)
- “An Ensemble Approach for Diabetes Prediction using Machine Learning” is accepted in Springer Conference.